

Multi-ancestry fine-mapping at scale

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Abstract

Genome-wide association studies have allowed researchers to identify millions of single-variant associations to common disease and measurement traits. These associations carry substantial value for explaining biological processes. However, associated variants are correlated through linkage disequilibrium. An association signal at a locus therefore does not by itself identify which correlated variant influences the phenotype. Bayesian fine-mapping methods quantify this uncertainty by assigning each variant at a locus a posterior inclusion probability — the probability that the variant itself is causal, conditioned on the observed association and the local LD structure. The variants at a locus are aggregated into a credible set that captures the true driver of the signal with a defined posterior probability. These methods were originally established for single-ancestry resolution, and already run at scale within the Open Targets Platform. Variants that tag the true signal through linkage disequilibrium, without driving it, show marginally different effects across populations due to differences in LD structure. A tag variant statistically indistinguishable from the true driver in one ancestry can be distinguishable in another. Multi-ancestry fine-mapping methods exploit the differences in LD by integrating data from multiple populations. The growth of large trans-ethnic biobanks producing GWAS for the same trait across ancestries allows multiple ancestries to be leveraged to improve fine-mapping accuracy for the underlying variants shared across ancestries. Grouping GWAS Catalog studies in Open Targets release 26.06 by shared trait resulted in 7,490 trait-sets with multiple single-ancestry studies. We are developing a scalable multi-ancestry fine-mapping workflow that integrates GWAS summary statistics, with ancestry-matched linkage disequilibrium reference panels from gnomAD and Pan-UKBB. The workflow is implemented using Nextflow on top of the Open Targets Gentropy framework, and is designed to run in-house at biobank scale. The pipeline resolves the Gentropy study index. For each trait set, pipeline determines which combinations of studies are eligible for multi-ancestry fine-mapping. The pipeline routes each eligible combination to one or more applicable methods. Studies assigned to an execution route are then clumped. Loci and their matched LD are passed through a shared Gentropy input interface. The method itself is agnostic and selectable, spanning Multi-SuSiE as proof of concept, with future work streamlined to integrate other methods. Method output is normalised back through a shared Gentropy output interface into the standard credible-set format, so that it can be reused efficiently by other post-GWAS analysis methods.

This Nextflow multi-ancestry fine-mapping pipeline will allow Open Targets to produce more accurate credible sets at scale. Built on open-source methods and uniform access to external LD reference panels, it will support reuse by other groups working with multi-ancestry GWAS data.